

Interacting Multipoles of Rank 1 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}^A\Phi_{\beta\gamma\delta\epsilon}^B = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} -945 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\alpha, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ +105 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \end{array} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}^A\Phi_{\beta\gamma\delta\epsilon}^B = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{aligned} &-945 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\delta, \delta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\delta\delta}^B \\ &+105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\gamma\delta}^B \\ &+105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\gamma\gamma\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\ &+105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\beta\gamma\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\beta\delta\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\ &+105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\gamma\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\delta\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\gamma\delta\epsilon}^B \\ &-15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ &-15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\ &-15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\ &-15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\ &-15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\epsilon}^B \\ &-15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\epsilon}^B \\ &-15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Phi_{\alpha\beta\beta\epsilon}^B \end{aligned} \right] \quad (2a)$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{aligned}
& -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\delta\delta}^B \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\gamma\delta}^B \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\gamma\epsilon}^B \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\epsilon}^B \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\delta\epsilon}^B \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\epsilon}^B \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\delta\epsilon}^B \\
& +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\alpha\gamma\delta\epsilon}^B \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\delta}^B \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\epsilon}^B \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\epsilon}^B \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\epsilon}^B
\end{aligned} \right] \quad (2b)
\end{aligned}$$

$$\begin{aligned} & \xrightarrow{\text{reduced indices}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{aligned} & -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\delta\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\ & +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\ & -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \end{aligned} \right] \end{aligned} \quad (2c)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{aligned} &-945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\delta\delta}^B \\ &+210 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\gamma\gamma\delta}^B \\ &+315 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\ &+420 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\ &-45 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ &-45 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \\ &-135 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^A \cdot \Phi_{\alpha\beta\beta\gamma}^B \end{aligned} \right] \quad (2d)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}^A\Phi_{\beta\gamma\delta\epsilon}^B = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{l} -945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Phi_{zzzz}^B \\ +105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Phi_{zz\delta\delta}^B \\ +210 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Phi_{zz\gamma\gamma}^B \\ +315 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Phi_{zz\beta\beta}^B \\ +420 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_{\alpha}^A \cdot \Phi_{zzz\alpha}^B \\ -45 \cdot R_z^4 \cdot R_z \cdot \mu_z^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ -45 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha}^A \cdot \Phi_{z\alpha\gamma\gamma}^B \\ -135 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha}^A \cdot \Phi_{z\alpha\beta\beta}^B \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} +420 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{zzz\alpha}^B - 45 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{z\alpha\gamma\gamma}^B \\ -135 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{z\alpha\beta\beta}^B \end{bmatrix} \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[\begin{array}{c} +420 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{zzz\alpha}^B - 45 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{z\alpha\beta\beta}^B \\ -135 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{z\alpha\beta\beta}^B \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{105} \cdot \left[+420 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{zzz\alpha}^B - 180 \cdot R_z^5 \cdot \mu_\alpha^A \cdot \Phi_{z\alpha\beta\beta}^B \right] \quad (3d)$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma\delta\epsilon} \mu_\alpha^A \Phi_{\beta\gamma\delta\epsilon}^B = R_z^{-11} \cdot -\frac{1}{105} \cdot \left[+420 \cdot R_z^5 \cdot \mu_y^A \cdot \Phi_{yzzz}^B - 180 \cdot R_z^5 \cdot \mu_y^A \cdot \Phi_{yz\beta\beta}^B \right] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{105} \cdot [+0] \quad (4b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon} \mu_\alpha^A \Phi_{\beta\gamma\delta\epsilon}^B = +0 \quad (5)$$